ShadowMem — Step 1 Closure Verification

Formal Audit Report: ERR-09 Resolution & server.py Retirement

Target Codebase: C:\Users\Himesh\Desktop\ShadowMem | **Audit Date:** October 06, 2026 - 21:40 | **Status:** Verification & Quality Assurance Audit

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1. Executive Verdict: Is Step 1 100% Complete?

VERDICT: 95% COMPLETE — CORE ENGINEERING PASS, DOCUMENTATION CONTRADICTIONS IDENTIFIED.
The core technical migration is a complete success: app.py is established as the single authoritative server, real-time memory map streaming is live, scan diffing is operational, input validation is active on /simulate/alert, and server.py was safely isolated into tests/mocks/server.py with its port rebound to 8082 to eliminate runtime port collisions. However, Step 1 closure is **not yet 100% complete** because **make_pdf.py was left half-updated and contains 6 internal contradictions** that still describe server.py as an active demo stack conflicting on port 8000.

2. Full Pass/Fail Verification Matrix (7 Key Checks)

Check # / Scope	Status	File & Line Reference	Concrete Findings & Runtime Verification Evidence
Check 1 **tests/mocks/server.py Port Rebinding**	**PASS**	`tests/mocks/server.py` `Lines 32 & 290`	• Line 32: Updated to uvicorn server:app --reload --port 8082. • Line 290: Updated to uvicorn.run(app, host='0.0.0.0', port=8082). • Line 29 added clarifying comment: *'(from tests/mocks/, standalone, for index.html testing only)'*. • Verified via git diff: **Zero other lines or logic altered**.

Check 2
PDF Scripts Text & Contradiction Check
	FAIL (Contradictions)	`make_pdf.py`
		`Lines 222, 316, 326, 445, 465, 540, 559, 563`
		`make_summary_pdf.py`
		`Lines 114, 181`
			• **make_summary_pdf.py: PASS.** Lines 114 and 181 correctly state WebSocket streaming is in app.py and server.py is a mock fixture.
			• **make_pdf.py: FAIL (Half-Updated).** While Section 3.2 (L316) and fact row (L559) were updated, **Lines 222, 326-332, 445, 463-467, 540, and 563 directly contradict this**, asserting the two stacks are unmerged and still conflict on port 8000.

Check 3 **app.py Startup & Clean Imports**	**PASS**	`app.py` `Lines 1-60, 257-270`	• Direct Python import executed cleanly: from app import app. • Zero import errors or lingering references from the server.py move. • Exactly 13 routes registered; programmatic route inspection confirmed **zero duplicate route registrations**.

Check 4
/api/scan Real Data Execution
	PASS	`app.py`
		`Lines 272-343`
		`memoryscanner.py`
		`Lines 90-174`
			• api_scan(BackgroundTasks()) executed across all active system processes.
			• Returned HTTP 200 with 133 real Windows processes inspected.
			• Summary verified: total_scanned: 133, anomalies: 2, malicious: 2, clean: 94, jit: 37, posture: 'Threat Detected'.

Check 5 **/ws/alerts Real Diffs & Memory Map**	**PASS**	`app.py` `Lines 887-985, 994-1065, 1068-1103`	• Dead sleep(60) loop replaced with _scan_ticker_loop() and receive_text(). • **Live Diffs Verified:** Caught 0 changes on identical scan, 1 change on new threat, 1 change on Suspicious→Malicious escalation. • **Real Memory Map Verified:** _build_memory_map() extracted 4 real MEM_PRIVATE executable regions from PID 22704 (Freebuff.exe) with real 64-bit addresses (e.g. 0x000002159A1D9000).
Check 6 **/simulate/alert Input Validation**	**PASS**	`app.py` `Lines 1112-1208`	• Accepts typed SimulateAlertRequest(scenario, severity). • Live tested: cobalt_strike_... returned T1620 frame; process_hollowing returned T1055.012 frame; dll_injection returned T1055.001 frame. • Rejections confirmed: Invalid scenario fake_attack → HTTP 400; Invalid severity ultra → HTTP 400; Missing severity → HTTP 422.

Check 7
Full Codebase Live Server Reference Scan
	PARTIAL (Confined to PDF)	`malware_simulator.py`
		`README_sandbox.md`
		`index.html`
		`launch_shadowmem.bat`
		`make_pdf.py`
			• **Production & Test Code: PASS.** launch_shadowmem.bat, malware_simulator.py (L38), README_sandbox.md (L338), and index.html (L2) all cleanly updated.
			• **Documentation Scripts: FAIL.** make_pdf.py still contains 6 references describing server.py as a live demo server running on port 8000.

3. Deep-Dive: Specific Contradictions in make_pdf.py

While freebuff AI updated lines 316 and 559 in make_pdf.py, it failed to update the rest of the file. The following exact lines in make_pdf.py directly contradict the retirement of server.py:

- **Line 222 (Section 1):** 'There are two parallel stacks in the repo: a REAL stack ... and a DEMO stack (server.py + index.html ...). They are not yet merged - that is the single biggest architectural TODO (Section 6).' → *Contradicts line 316 which states streaming lives in app.py.*
- **Lines 326–332 (Section 3.3):** '3.3 Intended (not yet built) unified architecture ... merging them is the top structural TODO (Section 6, item A).' → *Describes unified architecture as not yet built.*
- **Line 445 (Section 5 Inventory):** ['Synthetic streaming demo', 'server.py + index.html: WebSocket telemetry/alert/memory-map protocol...'] → *Lists server.py under active achieved demo modules.*
- **Lines 463–467 (Section 6 Backlog):** ['P0', 'B. Unify the two backends', 'app.py ... and server.py ... are separate apps that even conflict on port 8000. Merge: add /ws/alerts to app.py ... retire or clearly demote server.py to demo-only.'] → *Maintains unifying the backends as an open P0 backlog item.*
- **Line 540 (Quick-Run Table):** ['Mock demo stack', 'cd tests/mocks && python -m uvicorn server:app --port 8000...'] → *Directly instructs running server.py on port 8000 instead of 8082!*
- **Line 563 (Quick-Reference Fact Table):** ['Ports', 'Both stacks default 8000 - conflict to resolve during unification (6B)'] → *States that both stacks default to port 8000.*

4. Actionable Rectification Steps to Reach 100% Closure

To bring Step 1 from 95% to 100% verified closure, the following edits in make_pdf.py are required:

1. **Update Section 1 (Line 222):** Reword to state that the demo WebSocket stack has been successfully unified into app.py and server.py has been retired to tests/mocks/.
2. **Retire Section 3.3 (Lines 326–332):** Update heading to *'3.3 Completed unified architecture'* and confirm app.py now handles real scan diffing and memory maps.
3. **Mark Backlog Item B Complete (Lines 463–467):** Move 'Unify the two backends' from the P0 Backlog into Section 5 (Achieved capabilities).
4. **Fix Quick-Run Command (Line 540):** Change --port 8000 to --port 8082 to match the updated mock server binding.
5. **Update Fact Table (Line 563):** Update ports entry to reflect app.py = 8000 (primary) | tests/mocks/server.py = 8082 (test mock only).

Conclusion: *All technical code modifications for Step 1 (Win32 extraction, memory mapping, live ticker diffs, FastAPI routes, Pydantic simulation bodies, and port 8082 isolation) are verified and functioning flawlessly. Once the 6 documentation text strings in make_pdf.py are harmonized, Step 1 can be formally certified at 100% completion.*